

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20	U04		MODULAR DBB	K15AM10	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.
30			MODULAR DBB	K15AM14	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0		
40		VAV	MODULAR DBB	K09AM14	3	3	MODULAR VALVE DB&B BALL TYPE 900# RF INC 825 INC 625 TRIM METAL LO FULL SIZE TYPE 2BV	0	0	817	USE ONLY FOR TIE IN
50			MODULAR DBB	M15AM10	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B1	ALL BOLTS ARE METRIC DIA.
60			MODULAR DBB	M15AM14	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
70		VAW	MODULAR DBB	M15EM10	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
80		VAW	MODULAR DBB	M09EM10	1	1	MODULAR VALVE DB&B BALL TYPE 900# X 1500# RF INC 825 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
90							***** BALL VALVES *****			SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
90		VB2									
100											
100			BALL	A15AM13	2	2	BALL 1500# RF 825 INCONEL 625 TRIM TRUNNION SOFT RB LO	0	0		
110			BALL	A09AM13	3	3	BALL 900# RF 825 INCONEL 625 TRIM TRUNNION SOFT RB LO	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
120			BALL	A09AR15	4	10	BALL 900# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT RB GO	0	0		
130	U04	817	BALL	A09CR15-K23	16	16	BALL 900# CF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT RB GO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
140			BALL	B15AM09	1	1.5	BALL 1500# RF 825 INCONEL 625 TRIM FLOATING SOFT FB LO	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
150		VA5	BALL	B15AM13	2	2	BALL 1500# RF 825 INCONEL 625 TRIM TRUNNION SOFT FB LO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
160	U04	VA5	BALL	B09AM15	3	3	BALL 900# RF 825 INC 625 TRIM TRUNNION SOFT FB GO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
170		VA5	BALL	B09AR15	4	10	BALL 900# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT FB GO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
180	U04	817	BALL	B09CR15-K23	16	16	BALL 900# CF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT FB GO	0	0	VQ2	QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS.
190							***** THROUGH CONDUIT GATE VALVES *****				
190		VQ2									
200											
200			GATE	W09AM72	3	3	GATE 900# RF INC 825 TRIM 625 THROUGH COND SOLID SLAB HW	0	0		
210			GATE	W09AR72	4	4	GATE 900# RF LTCS A352 LCC + 825 CLAD INC 625 TRIM THROUGH COND SOLID SLAB HW	0	0		
220			GATE	W09AR73	6	10	GATE 900# RF LTCS A352 LCC + 825 CLAD INC 625 TRIM THROUGH COND SOLID SLAB GO	0	0		
230	U04	817	GATE	W09CR73	16	16	GATE 900# CF LTCS A352 LCC + 825 CLAD INC 625 TRIM THROUGH COND SOLID SLAB GO	0	0		
240							***** GLOBE VALVES *****				
240											
250											
250			GLOBE	L15AM43	1	2	GLOBE 1500# RF INC 825 INC TRIM 625 OS&Y HW	0	0		
260			GLOBE	L09AM43	3	3	GLOBE 900# RF INC 825 INC TRIM 625 OS&Y HW	0	0		
270			GLOBE	L09AR44	4	8	GLOBE 900# RF LTCS A352 LCC + 825 CLAD INC 625 TRIM OS&Y GO	0	0		
280							***** CHECK VALVES *****				
280											
290											
290		VC1	CHECK	C15AM51	1	1.5	CHECK PISTON 1500# RF INC 825 TRIM 625	0	0		
300	U04		CHECK	C15AM46	2	2	CHECK DUAL WAFER LUG 1500# RF INC 825 TRIM 625 THROU BOLTED	0	0		
310	U04		CHECK	C09AM46	3	3	CHECK DUAL WAFER LUG 900# RF INC 825 TRIM 625 THROU BOLTED	0	0		
320	U04		CHECK	C09AR46	4	8	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC + 825 CLAD TRIM INC 625 THROU BOLTED	0	0		
330	U04		CHECK	C09AR53	10	10	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC + 825 CLAD TRIM 625	0	0		
340	U04	817	CHECK	C09CR53-K23	16	16	CHECK DUAL DBL FLG 900# CF LTCS A352 LCC + 825 CLAD INC 625 TRIM	0	0		
350							***** PIPE *****				
350											
360		812	PIPE	5358511	1	1.5	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0		
370			PIPE	5358513	2	4	PIPE SCH XXS SMLS LTCS A333 GR.6 BE	0	0		
380			PIPE	5358515	6	6	PIPE SCH 160 SMLS LTCS A333 GR.6 BE	0	0		
390			PIPE	5358524	8	16	PIPE SCH 140 SMLS LTCS A333 GR.6 BE	0	0		
400							***** FITTINGS *****				
400											
410											
410			ELBOW	5392870	1	4	ELL 90 DEG LR XXS LTCS A420 WPL6	0	0		
420			ELBOW	5392816	6	6	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0		
430			ELBOW	5392762	8	16	ELL BW 90 DEG LR SCH 140 LTCS A420 WPL6	0	0		
440			ELBOW	5392876	1	4	ELL 45 DEG LR XXS LTCS A420 WPL6	0	0		
450			ELBOW	5392822	6	6	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6	0	0		
460			ELBOW	5392768	8	16	ELL BW 45 DEG LR SCH 140 LTCS A420 WPL6	0	0		
470											
470			TEE	5392879	1	4	TEE SCH XXS LTCS A420 WPL6	0	0		
480			TEE	5392825	6	6	TEE BW SCH 160 LTCS A420 WPL6	0	0		
490			TEE	5392771	8	16	TEE BW SCH 140 LTCS A420 WPL6	0	0		
500			TEE RED	5392877	1	3	TEE RED SCH XXS LTCS A420 WPL6	1.5	4		
510			TEE RED	5393073	3	4	TEE RED SCH 160/XXS LTCS A420 WPL6	6	6		
520			TEE RED	5393074	4	4	TEE RED SCH 140/XXS LTCS A420 WPL6	8	10		
530			TEE RED	5393075	6	6	TEE RED SCH 140/160 LTCS A420 WPL6	8	16		
540			TEE RED	5392770	8	14	TEE RED SCH 140 LTCS A420 WPL6	10	16		
550											

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 6 MM	SOUR SERVICE - NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 900 RF TEMPERATURE RANGE: -40 / 200 °C	12/16/2024	BY OZAC	U04	Page 1 of 3	900K23

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
550	U04		CAP	5392880	1	4	CAP BW SCH XXS LTCS A420 WPL6	0	0	
560			CAP	5392826	6	6	CAP BW SCH 160 LTCS A420 WPL6	0	0	
570			CAP	5392772	8	16	CAP SCH 140 LTCS A420 WPL6	0	0	
580										
580			RED CONC	5392881	1	3	REDUCER CONC SCH XXS LTCS A420 WPL6	1.5	4	
590			RED CONC	5395283	3	4	REDUCER CONC SCH 160/XXS LTCS A420 WPL6	6	6	
600			RED CONC	5395284	4	4	REDUCER CONC SCH 140/XXS LTCS A420 WPL6	8	10	
610			RED CONC	5395285	6	6	REDUCER CONC SCH 140/160 LTCS A420 WPL6	8	14	
620			RED CONC	5392773	8	14	REDUCER CONC SCH 140 LTCS A420 WPL6	10	16	
630			RED ECC	5392882	1	3	REDUCER ECC SCH XXS LTCS A420 WPL6	1.5	4	
640			RED ECC	5395286	3	4	REDUCER ECC SCH 160/XXS LTCS A420 WPL6	6	6	
650			RED ECC	5395287	4	4	REDUCER ECC SCH 140/XXS LTCS A420 WPL6	8	10	
660			RED ECC	5395288	6	6	REDUCER ECC SCH 140/160 LTCS A420 WPL6	8	14	
670			RED ECC	5392774	8	14	REDUCER ECC SCH 140 LTCS A420 WPL6	10	16	
680										
680							***** OLETS *****			
690										
690	U04		OLETS	5504164	1	2	NIPOFLANGE 1500# RF XXS LTCS A350 LF2 CL.1	2	4	
700	U04		OLETS	5504165	0.75	2	NIPOFLANGE 1500# RF SCH 160/XXS LTCS A350 LF2 CL.1	6	6	
710	U04		OLETS	5504217	0.75	2	NIPOFLANGE 1500# RF SCH 140/XXS LTCS A350 LF2 CL.1	8	16	
720	U04	UJ2	OLETS	5504169	1.5	1.5	NIPOFLANGE 1500# RF XXS LTCS A350 LF2 CL.1	4	4	
730	U04	UJ2	OLETS	5504167	1.5	1.5	NIPOFLANGE 1500# RF SCH 160/XXS LTCS A350 LF2 CL.1	6	6	
740	U04	UJ2	OLETS	5504218	1.5	1.5	NIPOFLANGE 1500# RF SCH 140/XXS LTCS A350 LF2 CL.1	8	16	
750										
750	U04		OLETS	5505197	1	1	WELDOLET SCH XXS LTCS A350 LF2 CL.1	3	4	
760	U04		OLETS	5505396	1	2	WELDOLET SCH 160/SCH XXS LTCS A350 LF2 CL.1	6	6	
770	U04		OLETS	5505181	1	4	WELDOLET SCH 140/XXS LTCS A350 LF2 CL.1	8	16	
780										
780	U04	UJ2	ELBOFLG	5516077	1.5	1.5	ELBOFLANGE 1500# RF XXS LTCS A350 LF2 CL.1	4	4	
790	U04	UJ2	ELBOFLG	5516078	1.5	1.5	ELBOFLANGE 1500# RF SCH 160/XXS LTCS A350 LF2 CL.1	6	6	
800	U04	UJ2	ELBOFLG	5516763	1.5	1.5	ELBOFLANGE 1500# RF SCH 140/XXS LTCS A350 LF2 CL.1	8	16	
810										
810		816					***** FLANGES *****			
820										
820	U04	FA8	TEE RED	5599704	0.5	0.5	FLG SCRDR (FNPT) REDUCING 1500# RF LTCS A350 LF2 CL.1	0.75	1.5	
830										
830			FLANGE	5851205	0.75	0.75	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
840			FLANGE	5851205	2	2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
850										
850	U04		FLANGE	5557505	1	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XXS BORE	0	0	
860	U04		FLANGE	5557146	3	4	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH XXS BORE	0	0	
870	U04		FLANGE	5557145	6	6	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 160	0	0	
880	U04		FLANGE	5557717	8	10	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 140 BORE	0	0	
890	U04			5557717	12	16	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 140 BORE	0	0	
900	U04	817	FLANGE	5619952	16	16	FLG COMPACT 900# RJ LTCS A350 LF2 CL.1 SCH 140 BORE	0	0	
910	U04		BLIND	5615616	0.5	2	FLG BLIND 1500# RF LTCS A350 LF2 CL.1	0	0	
920	U04		BLIND	5615605	3	10	FLG BLIND 900# RF LTCS A350 LF2 CL.1	0	0	
930	U04		BLIND	5615605	12	16	FLG BLIND 900# RF LTCS A350 LF2 CL.1	0	0	
940	U04	817	BLIND	5619774	16	16	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 CL.1 LINE CLASS 900K23	0	0	
950										
950							***** ORIFICE FLANGES *****			
960										
960	U04		ORIFICE FLG	5624364	2	2	WN ORIF SET 1500# RF LTCS A350 LF2 CL.1 SCH XXS BORE	0	0	
970	U04		ORIFICE FLG	5624365	3	4	WN ORIF SET 900# RF LTCS A350 LF2 CL.1 SCH XXS BORE	0	0	
980			ORIFICE FLG	5624537	6	6	WN ORIF SET 900# RF LTCS A350 SCH 160 BORE	0	0	
990	U04		ORIFICE FLG	5624366	8	16	WN ORIF SET 900# RF LTCS A350 LF2 CL.1 SCH 140 BORE	0	0	
1000										
1000							***** PIPE NIPPLES *****			
1010										
1010		812		0			NIPPLE BBE - CUT FROM PIPE			
1020										
1020							***** GASKETS *****			
1030										
1030			GASKET	5672251	0.5	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1040			GASKET	5673117	3	16	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1050	U04	817	GASKET	5673444	12	16	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 900K23	0	0	
1060										
1060		B1					***** BOLTING *****			
1070										
1070	U04		BOLTING	5675770	0.5	16	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1080										
1080		SB6	BOLTING	5675675	10	36	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1090										

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		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 6 MM	SOUR SERVICE - NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 900 RF TEMPERATURE RANGE: -40 / 200 °C	12/16/2024	BY OZAC	U04	Page 2 of 3	900K23

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1090							***** BLINDS & SPACERS *****			
1100			SPEC BLIND	5749595	0.75	2	SPECTACLE BLIND 1500# RF LTCS A516 GR.70	0	0	
1110			SPEC BLIND	5749596	3	8	SPECTACLE BLIND 900# RF LTCS A516 GR.70	0	0	
1120										
1120			PADDLE	5750102	10	10	PADDLE BLANK 900# RF LTCS A516 GR.70	0	0	
1130			PADDLE	5750102	12	16	PADDLE BLANK 900# RF LTCS A516 GR.70	0	0	
1140			PADDLE	5750202	10	10	PADDLE SPACER 900# RF LTCS A516 GR.70	0	0	
1150			PADDLE	5750202	12	16	PADDLE SPACER 900# RF LTCS A516 GR.70	0	0	
1160	U04	817	PADDLE	5750221	12	16	PADDLE BLANK COMPACT 900# RJ LTCS A516 GR.70	0	0	
1170	U04	817	PADDLE	5750225	12	16	PADDLE SPACER COMPACT 900# RJ LTCS A516 GR.70	0	0	
1180										

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